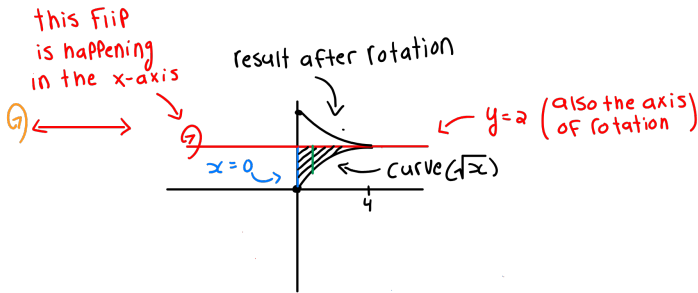


Sketching the region of typical disk/washer



$$y = \sqrt{x}, \quad x = 0, \quad y = 2$$

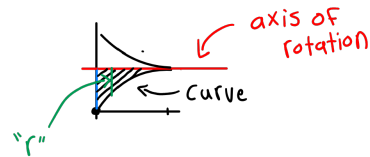
about the line $y = 2$

- about the line tells you where to rotate it "axis of rotation"

$$\text{area} = \int_0^4 \pi r^2 dx$$

- this is flipped over the x-axis so we use "dx" if it was flipped over the y-axis we would use "dy" so because it flipped in the x-axis the integral limit would go from 0 to 4

the "r" which stands for radius goes from the curve to the axis of rotation



For anything being flipped over the x-axis the equation for $r = \text{top} - \text{bottom}$

$$r = \text{top} - \text{bottom} \Rightarrow \int_0^4 \pi (2 - \sqrt{x})^2 dx$$

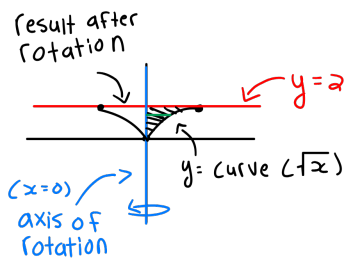
$$= 2 - \sqrt{x}$$

$$(2 - \sqrt{x})(2 - \sqrt{x}) \Rightarrow \pi \int_0^4 4 - 4\sqrt{x} + x dx$$

$$4 - 4\sqrt{x} + x$$

$$\pi \left[4x - \frac{8}{3} x^{3/2} + \frac{x^2}{2} \right]_0^4 \Rightarrow \pi \left[4(4) - \frac{8}{3} (4)^{3/2} + \frac{(4)^2}{2} \right]$$

$$= \boxed{\frac{8}{3} \pi}$$



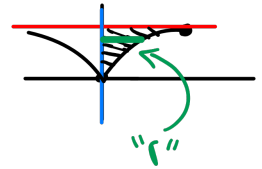
$$y = \sqrt{x}, \quad y = 2, \quad x = 0$$

about the y-axis

$$\text{area} = \int_0^2 \pi r^2 dy$$

- this is flipped over the y-axis so it will have a "dy" when its flipped over the y-axis the "r" is **right-left** and because this is flipped over the y-axis the integral limits goes from 0 to 2

r goes from the curve to the axis of rotation
use the original to plot "r"



$$r = \text{right} - \text{left} \Rightarrow \sqrt{x} - 0$$

$$y = \sqrt{x^2}$$

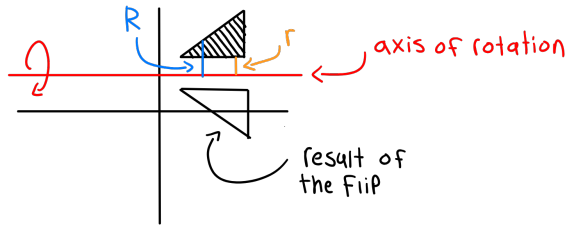
$$(y)^2 = (\sqrt{x})^2$$

$$x = y^2$$

- but here we face a problem we have "dy" we are integrating in terms of "y" but we have a " $\sqrt{x}$ " so need to get it in terms of y

$$r = y^2 - 0 \Rightarrow \int_0^2 \pi (y^2)^2 dy \Rightarrow \pi \int_0^2 y^4 dy \Rightarrow \pi \left[\frac{y^5}{5} \right]_0^2$$

$$\pi \left[\frac{(2)^5}{5} \right] \Rightarrow \boxed{\frac{32}{5} \pi}$$

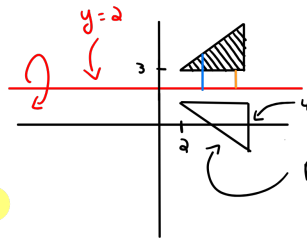


- this is when its flipped over the x -axis the same exact thing would be applied to previous examples but with two integrals

$$\int_a^b \pi R^2 dx + \int_a^b \pi r^2 dx$$

R for both " R " and " r " for rotation over the x -axis is top-bottom

For rotation over the y -axis the " R " and " r " is right-left



$$y = \sqrt{x}$$

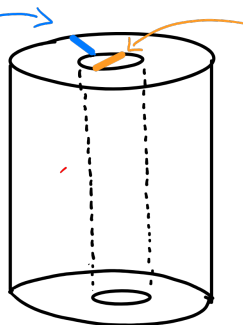
$$R = (\sqrt{x} - 2)$$

$$r = (3 - 2) = 1$$

$$\int_2^4 \pi (\sqrt{x} - 2)^2 dx + \int_2^4 \pi (1)^2 dx$$

when you have some like this you need to apply foil method do not do "u-sub"

large " R "
outside radius



small " r " is the
inside radius